Part 1: General Approach, Screening, and Follow-up 1. The 2017 guideline is an update of the "Seventh Report of the Joint National Committee on Prevention, Detection, Evaluation and Treatment of High Blood Pressure" (JNC 7), published in 2003. The 2017 guideline is a comprehensive guideline incorporating new information from studies regarding blood pressure (BP)-related risk of cardiovascular disease (CVD), ambulatory BP monitoring (ABPM), home BP monitoring (HBPM), BP thresholds to initiate antihypertensive drug treatment, BP goals of treatment, strategies to improve hypertension treatment and control, and various other important issues.

2. It is critical that health care providers follow the standards for accurate BP measurement. BP should be categorized as normal, elevated, or stages 1 or 2 hypertension to prevent and treat high BP. Normal BP is defined as hypertension stage 2 is ≥140 or ≥90 mm Hg. Prior to labeling a person with hypertension, it is important to use an average based on ≥2 readings obtained on ≥2 occasions to estimate the individual's level of BP. Out-of-office and self-monitoring of BP measurements are recommended to confirm the diagnosis of hypertension and for titration of BP-lowering medication, in conjunction with clinical interventions and telehealth counseling. Corresponding BPs based on site/methods are: office/clinic 140/90, HBPM 135/85, daytime ABPM 135/85, night-time ABPM 120/70, and 24-hour ABPM 130/80 mm Hg. In adults with an untreated systolic BP (SBP) >130 but 80 but

3. For an adult 45 years of age without hypertension, the 40-year risk for developing hypertension is 93% for African Americans, 92% for Hispanics, 86% for whites, and 84% for Chinese adults. In 2010, hypertension was the leading cause of death and disability adjusted life-years worldwide, and a greater contributor to events in women and African Americans compared with whites. Often overlooked, the risk for CVD increases in a log linear fashion; from SBP levels 180 mm Hg, and from DBP levels 105 mm Hg. A 20 mm Hg higher SBP and 10 mm Hg higher DBP are each associated with a doubling in the risk of death from stroke, heart disease, or other vascular disease. In persons ≥30 years of age, higher SBP and DBP are associated with increased risk for CVD, angina, myocardial infarction (MI), heart failure (HF), stroke, peripheral arterial disease, and abdominal aortic aneurysm. SBP has consistently been associated with increased CVD risk after adjustment for, or within strata of, SBP; this is not true for DBP.

4. It is important to screen for and manage other CVD risk factors in adults with hypertension: smoking, diabetes, dyslipidemia, excessive weight, low fitness, unhealthy diet, psychosocial stress, and sleep apnea. Basic testing for primary hypertension includes fasting blood glucose, complete blood cell count, lipids, basic metabolic panel, thyroid stimulating hormone, urinalysis, electrocardiogram with optional echocardiogram, uric acid, and urinary albumin-to-creatinine ratio.

5. Screening for secondary causes of hypertension is necessary for new-onset or uncontrolled hypertension in adults including drug-resistant (≥3 drugs), abrupt onset, age 30 years, excessive target organ damage (cerebral vascular disease, retinopathy, left ventricular hypertrophy, HF with preserved ejection fraction [HFpEF] and HF with reserved EF [HFrEF], coronary artery disease [CAD], chronic kidney disease [CKD], peripheral artery disease, albuminuria) or for onset of diastolic hypertension in older adults or in the presence of unprovoked or excessive hypokalemia. Screening includes testing for CKD, renovascular disease, primary aldosteronism, obstructive sleep apnea, drug-induced hypertension (nonsteroidal anti-inflammatory drugs, steroids/androgens, decongestants, caffeine, monoamine oxidase inhibitors), and alcohol-induced hypertension. If more specific clinical characteristics are present, screening for uncommon causes of secondary hypertension is indicated (pheochromocytoma, Cushing's syndrome, congenital adrenal hyperplasia, hypothyroidism, hyperthyroidism, and aortic coarctation). Physicians are advised to refer patients screening positive for these conditions to a clinician with specific expertise in the condition.

6. Nonpharmacologic interventions to reduce BP include: weight loss for overweight or obese patients with a heart healthy diet, sodium restriction, and potassium supplementation within the diet; and increased physical activity with a structured exercise program. Men should be limited to no more than 2 and women no more than 1 standard alcohol drink(s) per day. The usual impact of each lifestyle change is a 4-5 mm Hg decrease in SBP and 2-4 mm Hg decrease in DBP; but diet low in sodium, saturated fat, and total fat and increase in fruits, vegetables, and grains may decrease SBP by approximately 11 mm Hg.

7. The benefit of pharmacologic treatment for BP reduction is related to atherosclerotic CVD (ASCVD) risk. For a given magnitude reduction of BP, fewer individuals with high ASCVD risk would need to be treated to prevent a CVD event (i.e., lower number needed to treat) such as in older persons, those with coronary disease, diabetes, hyperlipidemia, smokers, and CKD. Use of BP-lowering medications is recommended for secondary prevention of recurrent CVD events in patients with clinical CVD and an average SBP ≥130 mm Hg or a DBP ≥80 mm Hg, or for primary prevention in adults with no history of CVD but with an estimated 10-year ASCVD risk of ≥10% and SBP ≥130 mm Hg or DBP ≥80 mm Hg. Use of BP-lowering medication is also recommended for primary prevention of CVD in adults with no history of CVD and with an estimated 10-year ASCVD risk <10% and a SBP ≥140 mm Hg or a DBP ≥90 mm Hg. The prevalence of hypertension is lower in women compared with men until about the fifth decade, but is higher later in life. While no randomized controlled trials have been powered to assess outcome specifically in women (e.g., SPRINT), other than special recommendations for management of hypertension during pregnancy, there is no evidence that the BP threshold for initiating drug treatment, the treatment target, the choice of initial antihypertensive medication, or the combination of medications for lowering BP differs for women compared with men. For adults with confirmed hypertension and known CVD or 10-year ASCVD event risk of 10% or higher, a BP target of <130/80 mm Hg is recommended. For adults with confirmed hypertension, but without additional markers of increased CVD risk, a BP target of <130/80 mm Hg is recommended as reasonable.

8. Follow-up: In low-risk adults with elevated BP or stage 1 hypertension with low ASCVD risk, BP should be repeated after 3-6 months of nonpharmacologic therapy. Adults with stage 1 hypertension and high ASCVD risk (≥10% 10-year ASCVD risk) should be managed with both nonpharmacologic and antihypertensive drug therapy with repeat BP in 1 month. Adults with stage 2 hypertension should be evaluated by a primary care provider within 1 month of initial diagnosis, and be treated with a combination of nonpharmacologic therapy and 2 antihypertensive drugs of different classes with repeat BP evaluation in 1 month. For adults with a very high average BP (e.g., ≥160 mm Hg or DBP ≥100 mm Hg), prompt evaluation and drug treatment followed by careful monitoring and upward dose adjustment is recommended.